E26: A Conditional L^2 Estimate for the Regularized Curvature Operator

Abstract

For $\varepsilon > 0$ and $\alpha \geq 0$, consider the curvature expression

$$\Theta_\varepsilon[\psi] = -\frac{\Delta\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}}.$$

We prove the conditional L^2 bound for the historical additive-denominator version of E26 under a precise, uniform lower bound on the modulus of the denominator. We also establish two explicit sufficient conditions and exhibit a smooth counterexample to the unconditional version of the estimate. In particular, $\varepsilon > 0$ alone does not prevent denominator zeros for signed real or complex fields.

1 Definitions and main result

Let $\Omega \subseteq \mathbb{R}^d$ be open, with $d \geq 1$. Fix $\varepsilon > 0$ and $\alpha \geq 0$ and take $\psi \in H^2(\Omega; \mathbb{C})$. Define

$$D_\varepsilon[\psi](x) := \psi(x) + \varepsilon e^{-\alpha|\psi(x)|^2}. \tag{1}$$

Where $D_\varepsilon[\psi] \neq 0$, let

$$\Theta_\varepsilon[\psi](x) := -\frac{\Delta\psi(x)}{D_\varepsilon[\psi](x)}. \tag{2}$$

Pointwise assertions about Sobolev representatives are always understood almost everywhere.

Theorem 1 (E26: conditional L^2 estimate)**.** *Assume that there is a constant $\delta > 0$ for which*

$$\operatorname*{ess\,inf}_{x\in\Omega} \left|D_\varepsilon[\psi](x)\right| \geq \delta. \tag{3}$$

Then $\Theta_\varepsilon[\psi] \in L^2(\Omega; \mathbb{C})$ and

$$\boxed{\left\|\Theta_\varepsilon[\psi]\right\|_{L^2(\Omega)} \leq \frac{1}{\delta} \left\|\Delta\psi\right\|_{L^2(\Omega)}.} \tag{4}$$

The estimate is valid for every $d \geq 1$; no additional boundary condition is needed for this inequality.

Proof. Since $\psi \in H^2(\Omega; \mathbb{C})$, every weak second partial derivative of ψ belongs to $L^2(\Omega)$, and hence

$$\Delta\psi = \sum_{j=1}^{d} \partial_j^2 \psi \in L^2(\Omega; \mathbb{C}).$$

The expression $D_\varepsilon[\psi]$ is measurable. Hypothesis (3) implies that it is nonzero almost everywhere and that its reciprocal is essentially bounded:

$$\left\| \frac{1}{D_\varepsilon[\psi]} \right\|_{L^\infty(\Omega)} \leq \frac{1}{\delta}. \tag{5}$$

The product of an L^∞ function and an L^2 function belongs to L^2. In particular, by Hölder's inequality,

$$\begin{aligned}\|\Theta_\varepsilon[\psi]\|_{L^2(\Omega)} &= \left\| \frac{\Delta\psi}{D_\varepsilon[\psi]} \right\|_{L^2(\Omega)} \\ &\le \left\| \frac{1}{D_\varepsilon[\psi]} \right\|_{L^\infty(\Omega)} \|\Delta\psi\|_{L^2(\Omega)} \\ &\le \frac{1}{\delta} \|\Delta\psi\|_{L^2(\Omega)}.\end{aligned}$$

This proves (4). Equivalently, one can calculate directly:

$$\begin{aligned}\|\Theta_\varepsilon[\psi]\|_{L^2(\Omega)}^2 &= \int_\Omega \frac{|\Delta\psi(x)|^2}{|D_\varepsilon[\psi](x)|^2}\, dx \\ &\le \delta^{-2} \int_\Omega |\Delta\psi(x)|^2\, dx \\ &= \delta^{-2} \|\Delta\psi\|_{L^2(\Omega)}^2.\end{aligned}$$

Taking nonnegative square roots gives the same conclusion. □

2 Sufficient denominator bounds

Corollary 1 (Fields bounded away from zero)**.** *If $|\psi(x)| \ge m > \varepsilon$ almost everywhere on Ω, then*

$$\|\Theta_\varepsilon[\psi]\|_{L^2(\Omega)} \le \frac{1}{m-\varepsilon} \|\Delta\psi\|_{L^2(\Omega)}. \tag{6}$$

Proof. As $\alpha \ge 0$, one has $0 < \varepsilon e^{-\alpha|\psi|^2} \le \varepsilon$. The reverse triangle inequality therefore yields

$$|D_\varepsilon[\psi]| \ge |\psi| - \varepsilon e^{-\alpha|\psi|^2} \ge m - \varepsilon > 0.$$

Apply Theorem 1 with $\delta = m - \varepsilon$. □

Remark 1. On $\Omega = \mathbb{R}^d$, an L^2 function cannot satisfy $|\psi| \ge m > 0$ almost everywhere. Thus Corollary 1 is most naturally used on finite-measure domains. The main theorem itself applies equally well on unbounded domains whenever its hypothesis is satisfied.

Corollary 2 (Nonnegative real fields)**.** *Suppose $\psi \in H^2(\Omega;\mathbb{R})$ and $\psi \ge 0$ almost everywhere. For every fixed $\varepsilon > 0$ and $\alpha \ge 0$, set*

$$\delta_\varepsilon := \min\left\{ \frac{\varepsilon}{2},\, \varepsilon e^{-\alpha\varepsilon^2/4} \right\} > 0. \tag{7}$$

Then

$$\|\Theta_\varepsilon[\psi]\|_{L^2(\Omega)} \le \frac{1}{\delta_\varepsilon} \|\Delta\psi\|_{L^2(\Omega)}. \tag{8}$$

Proof. For almost every x, write $r = \psi(x) \ge 0$. If $r \ge \varepsilon/2$, then

$$r + \varepsilon e^{-\alpha r^2} \ge \varepsilon/2.$$

If $0 \le r < \varepsilon/2$, the assumption $\alpha \ge 0$ yields

$$r + \varepsilon e^{-\alpha r^2} \ge \varepsilon e^{-\alpha\varepsilon^2/4}.$$

Thus $D_\varepsilon[\psi] \ge \delta_\varepsilon$ almost everywhere. The assertion follows from Theorem 1. □

3 Counterexample without denominator separation

Counterexample 1 (Positive regularization is not enough). Choose $\Omega = (-1, 1)$, take any $\varepsilon > 0$, set $\alpha = 0$, and define the smooth real-valued function

$$\psi(x) = x^2 - \varepsilon.$$

Then $\psi \in H^2((-1, 1))$ and

$$D_\varepsilon[\psi](x) = x^2, \qquad \Delta\psi(x) = \psi''(x) = 2.$$

Although the denominator vanishes only at $x = 0$, the quotient is not square-integrable:

$$\Theta_\varepsilon[\psi](x) = -\frac{2}{x^2} \quad (x \neq 0), \qquad \int_{-1}^{1} |\Theta_\varepsilon[\psi](x)|^2 \, dx = 4 \int_{-1}^{1} |x|^{-4} \, dx = +\infty.$$

In contrast,

$$\|\Delta\psi\|^2_{L^2((-1,1))} = \int_{-1}^{1} 4 \, dx = 8 < \infty.$$

Consequently, the E26 estimate is *false* for unrestricted signed real fields (and hence for unrestricted complex fields) if one assumes only $\varepsilon > 0$ and $\psi \in H^2$.

4 Relationship to the corrected WCT regularization

The operator in (2) is the *historical* additive-denominator regularization. It must not be identified with the phase-covariant modulus-squared regularization used in M2+E17, which instead has the form

$$\widetilde{\Theta}_\varepsilon[\psi] = -(\Delta\psi) \, \frac{\overline{\psi}}{|\psi|^2 + \varepsilon^2 e^{-2\alpha|\psi|^2}}.$$

For fixed $\varepsilon > 0$ and $\alpha \geq 0$, the corrected multiplier

$$R_\varepsilon(z) := \frac{\overline{z}}{|z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}}$$

is bounded on $\mathbb{C}$. If $|z| \geq \varepsilon/2$, then $|R_\varepsilon(z)| \leq |z|^{-1} \leq 2/\varepsilon$. Otherwise,

$$|R_\varepsilon(z)| \leq \frac{|z| e^{2\alpha|z|^2}}{\varepsilon^2} \leq \frac{e^{\alpha\varepsilon^2/2}}{2\varepsilon}.$$

Thus

$$\|\widetilde{\Theta}_\varepsilon[\psi]\|_{L^2} \leq \frac{1}{\varepsilon} \max\left\{2, \frac{e^{\alpha\varepsilon^2/2}}{2}\right\} \|\Delta\psi\|_{L^2} \quad (\psi \in H^2(\Omega; \mathbb{C})).$$

This unconditional fixed-regularization estimate and E26 for the historical operator are different assertions. Neither gives a uniform $\varepsilon \downarrow 0$ bound without further hypotheses.

5 Scope and interpretation

Hypothesis (3) is a bound on the *modulus of the full denominator*. It does not follow from the mere positivity of ε. Nor may the constant δ be replaced by ε without additional restrictions on ψ.

Theorem 1 is a pointwise-in-ψ multiplication estimate, not an estimate for the difference of operators evaluated at two different fields. It establishes neither local Lipschitz continuity nor global existence or uniqueness for an associated evolution problem. Furthermore, the theorem supplies no uniform-in-ε estimate as $\varepsilon \downarrow 0$ unless a positive denominator bound independent of ε is separately established.